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Key Points:
- Landfast ice can be parametrized by adjusting sea ice strength
- Landfast ice has an influence on the Arctic ocean halocline stability
- The effects of the landfast ice vary regionally

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Landfast ice affects the stability of the Arctic halocline: Evidence from a numerical model

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Abstract Landfast ice covers large surface areas of the winter Siberian Seas. The immobile landfast ice cover inhibits divergent and convergent motion, hence dynamical sea ice growth and redistribution, decouples winter river plumes in coastal seas from the atmosphere, and positions polynyas at the landfast ice edge offshore. In spite of the potentially large effects, state-of-the-art numerical models usually do not represent landfast ice in its correct extent. A simple parametrization of landfast ice based on bathymetry and internal sea ice strength is introduced and its effects on the simulated Arctic Ocean are demonstrated. The simulations suggest that the Siberian landfast ice impacts the Arctic halocline stability through enhanced brine production in polynyas located closer to the shelf break and by redirecting river water to the Canadian Basin. These processes strengthen the halocline in the Canadian Basin, but erode its stability in the Makarov and Eurasian Basins.

1. Introduction

One of the dominant characteristics of the winter Arctic shelf seas is landfast ice (also landfast, fast, or shorefast ice), sea ice that is immobile and mechanically fastened to the coast or to the sea floor. As there is no compressive deformation, landfast ice grows only thermodynamically and rarely exceeds thicknesses of 1.5 m [*Romanov*, 2004]. It can extend a few (Beaufort Sea, Chukchi Sea, Western Laptev Sea) to several hundred kilometers from the coast into the ocean (Kara Sea, Eastern Laptev Sea, East Siberian Sea, see Figure 1). The mechanisms that determine the landfast ice formation, extent, and decay are not fully understood. To complicate things further, these mechanisms differ regionally. In the Chukchi and Beaufort Sea, the landfast ice edge is found at relatively shallow depth of 18 m [*Mahoney et al*., 2007], while on the Eurasian shelf this depth is between 25 and 30 m [*Dmitrenko et al*., 2005b; *Proshutinsky et al*., 2007]. In the Chukchi and Beaufort Sea, the landfast ice is immobilized behind a row of bottom reaching pressure ridges [*Mahoney et al*., 2007]. In the Kara Sea, the landfast ice is formed behind a row of small islands parallel to the coast [*Divine et al*., 2004]. At the landfast ice edge in the Laptev and East Siberian Seas, grounded pressure ridges are rare [*Reimnitz et al*., 1994; *Eicken et al*., 2005]. *Proshutinsky et al*. [2007] proposed that the landfast ice edge occurs where the warm intermediate Atlantic water reaches the surface after upwelling at the shelfbreak. *König Beatty and Holland* [2010] attributed the landfast ice extent to the mechanical properties of the sea ice. Strong freshwater and brackish sea ice [*Dethleff et al*., 1993; *Eicken et al*., 2005] formed in low-salinity shelf seas with high river water content might ground at wide and shallow sand banks [*Dethleff et al*., 1993; *Reimnitz et al*., 1994] and from there extend with long tongues toward the landfast ice edge. Tides in the Southeastern Laptev Sea are weak and the tidal amplitudes that do not exceed 10 cm [*Fofonova et al*., 2014] might not be large enough to deform the extensive landfast ice cover.

Especially in the Eurasian shelf seas, where landfast ice has the largest extent, it is thought to have three roles:

1. Landfast ice limits the sea ice thickness by preventing sea ice compression (e.g., pressure ridges) in convergent motion of sea ice. By the same token, it lowers the sea ice production by preventing sea ice divergence and lead formation. If these processes are not adequately represented in a numerical model, simulated thickness fields are not realistic [*Johnson et al*., 2012].

2. An immobile lid of the landfast ice effectively decouples the inner shelf from the atmosphere and affects the river water distribution. The large amount of the fresh river water that is discharged by the Siberian rivers (Ob, Lena, Yenisei) into the surface layer of the Eurasian shelf seas enhances the ocean stratification. The runoff of these rivers has a strong seasonal cycle with a peak in June and very moderate discharge during

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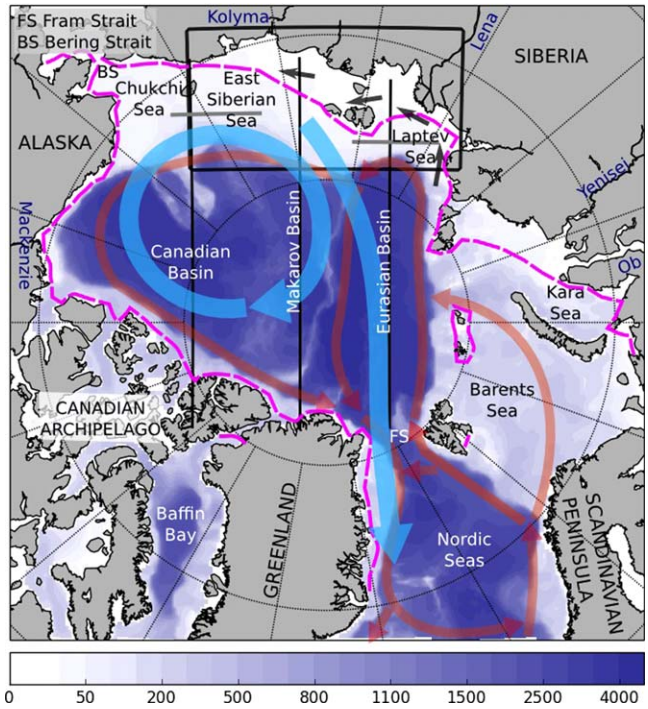

Figure 1. The Arctic Ocean and its marginal seas. The sea ice and surface circulation are schematically represented by light blue and mid depth circulation by red arrows (simplified from *Rudels et al*. [1994]), respectively. The landfast ice edge is depicted by the magenta dash line. The coastal current in the Laptev and East Siberian Seas is schematically represented by the black arrows. The black lines and box mark the sections and regions used in the model analysis.

the rest of the year. The landfast ice that covers the shelf only during the cold part of the year can still influence the river water distribution because large parts of the summer maximal discharge can be held back on the inner shelf by strong onshore winds until winter [*Dmitrenko et al*., 2005a].

3. The extent of landfast ice determines the areas of low sea ice concentration—so-called flaw polynyas (also flaw lead polynyas or flaw leads)—along the landfast ice edge during offshore wind conditions. The water in these polynyas is more saline, and consequently more brine is rejected during sea ice production than directly at the coast, where ocean salinities remain low due to fresh river water inflow. The brine formed in the winter shelf seas maintain, along with the cool water formed by winter convection north of the Barents Sea [*Rudels et al*., 1996], the Arctic halocline [*Aagaard et al*., 1981; *Martin and Cavalieri*, 1989; *Cavalieri and Martin*, 1994; *Winsor and Björk*, 2000]. The cold and saline water flows off the shelves and sinks along the shelf break where it feeds into the halocline layer (Figure 2), which decouples the cold Arctic surface layer from the warm intermediate-depth Atlantic layer. The surface layer stays cold with temperatures close to the freezing point due to a seasonal sea ice melting—freezing cycle and the layer is fresh due to a strong river runoff. In contrast, the ocean temperature at mid depth is up to several degrees above zero. This warm and saline Atlantic water enters the ocean through the Nordic Seas (Figure 1).

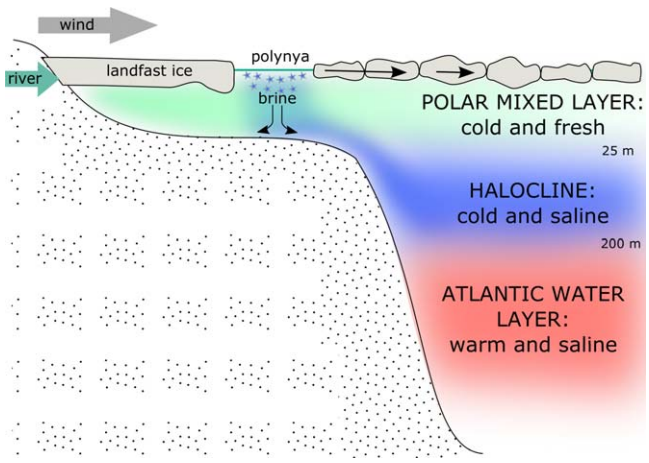

Figure 2. Arctic Ocean vertical stratification and winter processes maintaining the Arctic halocline.

Landfast ice has been found important for the accurate simulation of the sea surface height [*Proshutinsky et al*., 2007] and sea ice thickness *Johnson et al*. [2012]. Although landfast ice is also important for the processes maintaining the Arctic halocline, it is usually not properly represented in state-of-the-art sea ice-ocean models. In this study, we demonstrate the effects of the landfast ice for the Arctic halocline.

The outline of this paper is as follows. In section 2, we describe the numerical model for the sensitivity study. In section 2.1, we describe the details of the landfast ice

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parametrization. In sections 3, 4, and 5, we present and discuss our results. A summary of our findings and final remarks are given in section 6.

2. Model Setup

Our model is a regional coupled sea ice—ocean model based on the Massachusetts Institute of Technology General Circulation Model code—MITgcm [*Marshall et al*., 1997; *MITgcm Group*, 2014] with a model domain covering the Arctic Ocean, Nordic Seas, and northern North Atlantic. The horizontal resolution of is 1/4° (∼28 km) on a rotated grid with the grid equator passing through the geographical North Pole. The model has 36 vertical levels unevenly distributed in a way that the surface layer is well resolved at the cost of the poor resolution in the deep layer. The shelf bottom topography has realistic details that allow dense brine to flow downslope and off the shelf. Vertical mixing in the ocean is parameterized by a K-Profile Parameterization (KPP) scheme [*Large et al*., 1994] and tracers (temperature and salinity) are advected with an unconditionally stable seventh-order monotonicity preserving scheme [*Daru and Tenaud*, 2004] that requires no explicit diffusivity. The sea ice model is a dynamic-thermodynamic sea-ice model with a viscous-plastic rheology [*Losch et al*., 2010]. The model has a nonlinear free surface and sea ice that depresses the surface ocean layer according to its mass. We avoid numerical issues associated with too thin surface layers when ice gets very thick by using a rescaled vertical z*-coordinate [*Campin et al*., 2008] that distributes the excursion of the free sea surface between all vertical levels to the bottom. The same model setup has been used by *Itkin et al*. [2014] except that now the vertical background diffusivity has been lowered to 10^{-6} m^2/s as recommended by *Nguyen et al*. [2009] to achieve a better defined Arctic halocline.

The model is initialized by the PHC climatology [*Steele et al*., 2001] and has initially no sea ice cover. For a spin-up, we run the model for 30 years forced by the atmospheric climatology of the Coordinated Ocean Research Experiment (CORE) version 2 based on a reanalysis of the National Center for Atmospheric Research/National Centers for Environmental Prediction (NCAR/NCEP) [*Large and Yeager*, 2009]. Subsequently the model is driven from 1948 to 1978 by daily atmospheric data also provided by CORE. Our model experiments start in 1979 and continue until 2010. They are forced by the atmospheric reanalysis—The Climate Forecast System Reanalysis (NCEP–CFSR) [*Saha et al*., 2010]. Surface salinity in ice free regions is restored to a mean salinity field (PHC climatology) with a time scale of 180 days to avoid model drift. River runoff was treated like a surface volume flux and it was prescribed for the main Arctic rivers according to the Arctic Ocean Model Intercomparison Project (AOMIP, http://www.whoi.edu/projects/AOMIP/) protocol. Open boundaries are formulated following *Stevens* [1991]; they are located at 50°N in the Atlantic and just south of the Bering Strait. Temperature and salinity at the open boundaries are taken from the PHC climatology. The Barents Strait inflow is prescribed as 0.8 Sv and the stream function at the open boundary in the Atlantic Ocean is derived from a North Atlantic simulation [*Gerdes and Köberle*, 1995] and modified to balance the net volume flow in the model domain.

2.1. Landfast Ice Parameterization

The mechanisms that determine the landfast ice formation and extent differ from region to region and are not fully understood. There have been many attempts to model or parameterize landfast ice. *Lieser* [2004] developed a parameterization based on the ratio of sea ice thickness and total water column depth. For a specified threshold ratio, the grid cells assigned as landfast ice remained at rest and surface momentum flux into the ocean was set to zero. His approach in a 1/4° (28 km) model resulted in too thick landfast ice that even survived the summer. A similar procedure was used in higher horizontal resolution models (3–12 km) by *Johnson et al*. [2012] and *Rozman et al*. [2011]. Both studies use prescribed landfast ice areas obtained from bathymetric limits or observations. In the Kara Sea, where the landfast ice forms over deep waters behind a row of coastal islands, *Olason* [2012] successfully modeled landfast ice by adjusting the internal sea ice strength parameters in the viscous-plastic rheology of *Hibler* [1979].

Our model grid does not resolve small islands and shallow topographical features in the Siberian Seas, where the landfast ice might get grounded. Therefore, we designed a simplified and uniform parametrization based on *König Beatty and Holland* [2010] that takes into account water column depth and landfast ice internal strength. The latter is justified by low salinity surface water of the Kara, Laptev, and East Siberian Sea shelf. The ice formed in fresh or brackish water has a lower porosity and consequently higher internal strength than sea ice formed in the ocean water [e.g., *Timco and Weeks*, 2010]. In 1992, *Dethleff et al*. [1993]

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documented freshwater ice up to 100 km seaward off the Lena Delta. In 1999, the freshwater and brackish sea ice with salinity below 1 was confined to the shallow coastal waters adjacent to the eastern Lena Delta with water depths below 10 m, while 62% of the landfast ice further offshore in the southeastern Laptev Sea was formed from river water [*Eicken et al*., 2005].

In the widely used sea ice strength parametrization [e.g., *Hibler*, 1979; *Zhang and Hibler*, 1997], compressive strength P depends just on the sea ice thickness h and concentration A:

$$P = P^* h \ \exp\left(-C^*(1-A)\right), \tag{1}$$

where the empirical sea ice strength parameters, $P^* = 2750 N/m^2$ and $C^* = 20$ are constants.

The landfast ice parametrization used in this study takes into account the unresolved shallow topographical features inside the maximal landfast ice edge mark (25 m) and increased sea ice internal strength attributed to the lower sea ice salinity in the same area by setting the P^* to the double of the drift ice. In our model simulation with such a parametrization, landfast ice still fails under strong offshore wind at locations where the distance between the coastline and the 25 m isobath is large. To prevent this, we amended the sea ice rheology in the regions shallower than 25 m with sea ice tensile strength T following *König Beatty and Holland* [2010]:

$$\zeta = \frac{P+T}{2\Delta}, \tag{2}$$

$$\eta = \frac{P+T}{2\Delta e^2} = \zeta/e^2, \tag{3}$$

and

$$p = P - T, \tag{4}$$

where ζ is bulk viscosity, η is shear viscosity, $e = 2$ is eccentricity constant, p is pressure term, and $\Delta = \sqrt{(\dot{\epsilon}_{11}^2 + \dot{\epsilon}_{22}^2)(1+e^{-2}) + \frac{4}{e^2}\dot{\epsilon}_{12}^2 + 2\dot{\epsilon}_{11}\dot{\epsilon}_{22}(1-e^{-2})}$ with the strain rate tensor $\dot{\epsilon}_{ij} = 0.5(\partial_i u_j + \partial_j u_i)$. The original rheology by *Hibler* [1979] models sea ice with T nearly zero, but *König Beatty and Holland*'s formulation increases the T by moving the elliptical yield curve in the principal stress space into the I quadrant, when the water column is shallower than 25 m and leaves the curve unmodified otherwise. We choose $T = \frac{P}{2}$, which is consistent with previous estimates [*Tremblay and Hakakian*, 2006].

For the sensitivity study, we compare a control run (CTRL) and a landfast ice run (LF) that differ only in this additional landfast ice parameterization.

3. Impact of the Landfast Parameterization on the Sea Ice

In the Laptev and East Siberian Sea, the landfast ice cover forms in December, reaches its maximal extent in April, breaks up into fully drifting ice in May or June, and melts in summer. Starting in May, the atmospheric temperatures are too warm to have a significant amount of brine produced in the polynya. Hence, in this study, we define wintertime as the months December to April. In our model, landfast ice breaks up completely in early summer. At atmospheric temperatures above zero and when A drops below 1, sea ice looses its internal strength exponentially making P^* irrelevant, because P depends exponentially on A.

If we define coastal sea ice with drift speeds below 1 mm/s as landfast ice, then the parametrization in LF, compared to CTRL, leads to substantially larger areas of landfast ice during the annual maximal extent in April (Figures 3a and 3b). The simulated landfast ice is slow enough for polynyas to form at its seaward edge (Figures 3c and 3d). The 1 mm/s contour and polynyas in LF agree better with the 1997–2006 landfast ice edge produced by the Arctic and Antarctic Research Institute, Saint Petersburg, Russia (*Smolyanitsky et al*. [2007], hereafter AARI data set). In the Kara Sea, our parametrization has only a minor effect because the shelf sea is much deeper than the 25 m threshold we use in the parametrization and the small offshore islands there are located in areas where the surrounding seas exceed the depth of 150 m.

The effect of the parametrization is clearly visible in the sea ice thickness maps (Figure 4). LF sea ice is by up to 30 cm thinner in the regions affected by the landfast ice parametrization in the Kara, Laptev, East Siberian, Chukchi, and Beaufort Seas. The different pattern accurately matches the landfast ice extent from the

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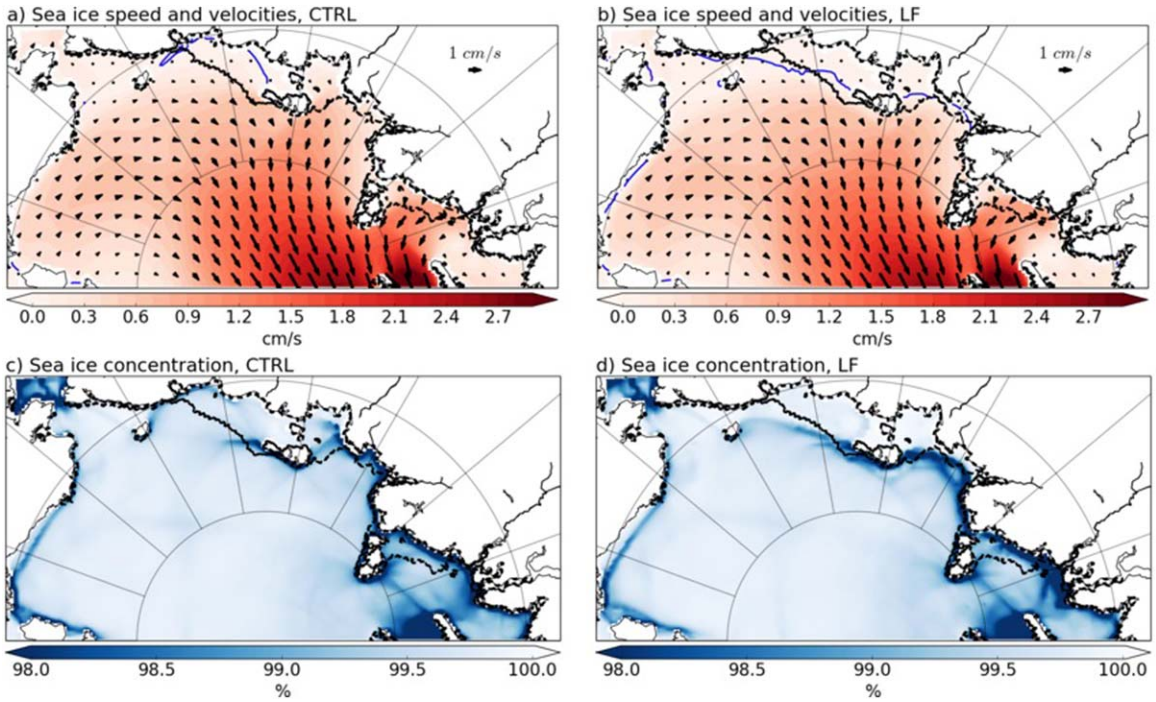

Figure 3. The effect of the landfast ice parametrization on the mean April (2000–2010) sea ice motion (a,b) and concentration (c,d). a,c—CTRL, b,d—LF. Speed 1 mm/s is contoured by blue line. Mean April landfast edge from the AARI data set (1997–2006) is depicted by black dashed line.

AARI data set (Figure 3). The largest negative differences are in the polynya at the landfast ice edge in the East Siberian and Laptev Seas, while in the coastal regions of the Laptev Sea differences are positive, reflecting the erroneous polynya location in the CTRL.

The thinner sea ice in the landfast ice area in LF is a consequence of the absence of ridging. The location of the thermodynamical growth maxima is also different between the runs (Figure 4): in both cases the highest growth rates occur in the polynyas. In LF, this is further offshore than in CTRL.

As a volume of water is thermodynamically turned into sea ice, salt is expelled from this volume and remains in the ocean. Because this process increases the salinity in the surface ocean, one can define a "virtual salt flux" $S_{flux}=FW_{flux}S$, where FW_{flux} is freshwater flux and S is the salinity of the surface water minus 4 where the salinity is equal or higher than 4. This approximates the constant sea ice salinity of 4 in our model. The resulting S_{flux} differences resemble the difference pattern in the thermodynamic growth (Figure 4). Because S_{flux} depends also on the sea surface salinity, it is relatively low in the East Siberian Sea although the sea ice production there is high. S_{flux} for the region is for every winter an order of magnitude higher and opposite in sign compared to the virtual salt flux generated by the sea surface salinity restoring (not shown). Salinity is only restored to climatology in the ice free part of the model grid cell.

The largest differences in the winter sea ice cover between the runs are found in the Laptev and East Siberian Seas (hereafter LESS). This is the area with the greatest landfast ice extent over the shallow shelf and hence our parametrization has the largest effect there. In time series, we compare the contributions of wintertime sea ice concentration, thickness, sea ice production, ocean surface salinity, and S_{flux} to wintertime dense shelf water production in both simulations (Figure 5). The mean sea ice concentration over the LESS is high and very similar in both simulations. This is not surprising as the differences between the runs are mainly in the positioning of the polynyas within the LESS. The sea ice concentrations are a little lower in CTRL as the drifting sea ice allows more small leads. On average, sea ice is 8 cm thinner in LF than in CTRL. This is a consequence of less ridging through convergent motion in the landfast ice area and less advection of sea ice into the region. In contrast, there is more thermodynamically grown sea ice in LF. While the polynyas in LF are shifted offshore into more saline sea water, a higher fraction of fresh river water reduces the salinity in the shelf seas. Toward the end of the runs, these counteracting effects lead to a similar surface salinity in polynyas in both runs. Still, higher sea ice production leads to higher S_{flux} in LF.

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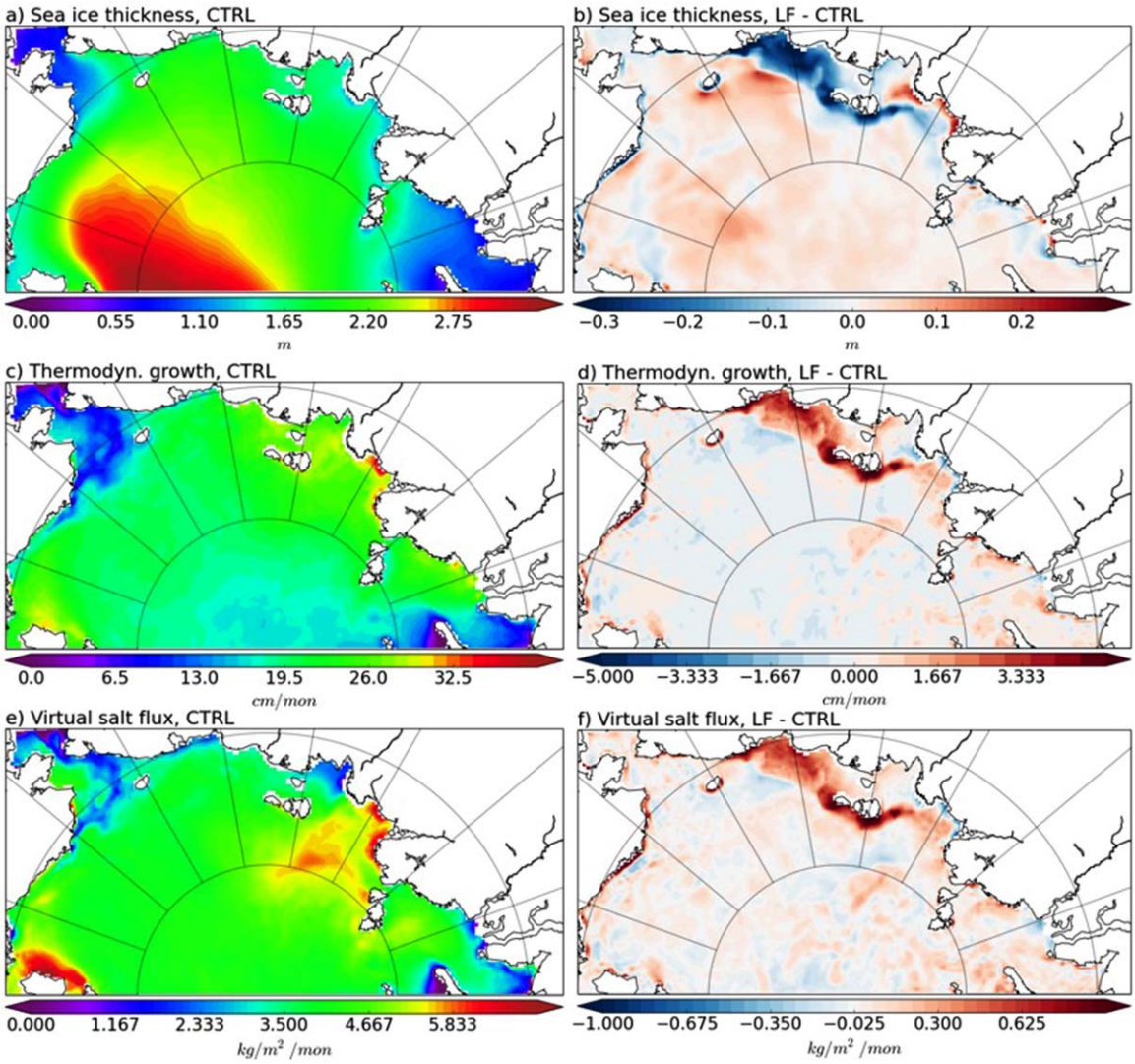

Figure 4. Mean winter (2000–2010) sea ice thickness, thermodynamical growth, and salt flux from the sea ice thermodynamical growth: (a) CTRL, (b) LF minus CTRL.

The differences in the sea ice production and S_{flux} between CTRL and LF reflect the differences between a simulation with a geographically limited representation of the landfast ice cover and a simulation which is faithful to the observed landfast ice extent. The runs can therefore be further compared to estimate the importance of the landfast ice for the Arctic Ocean halocline stability.

4. Impact of the Landfast Parameterization on the Arctic Halocline

In this section, we compare the end of winter (April) climatologies (2000–2010) because of the high seasonality of the mixed layer depth in the Arctic. First we examine the buoyancy, $b=-g\frac{\rho-\rho_0}{\rho_0}$ at the top of the halocline (25–30 m). The differences between LF and CTRL show regional effects of the landfast ice parametrization with less buoyancy in LF in the Canadian Basin and more buoyancy centered in the Makarov and Eurasian Basins (Figures 6a and 6b).

To understand the relevance of these differences to the water column stability, we examine three oceanographic sections; each section runs through one of the major Arctic basins: Canadian, Makarov, and Eurasian as shown on Figures 1 and 6. In the first section running from the Eurasian coast, over Wrangel Island, to Ellesmere Island (Figure 7), there is increased salinity by locally up to 0.6 at the halocline layer in LF (25–

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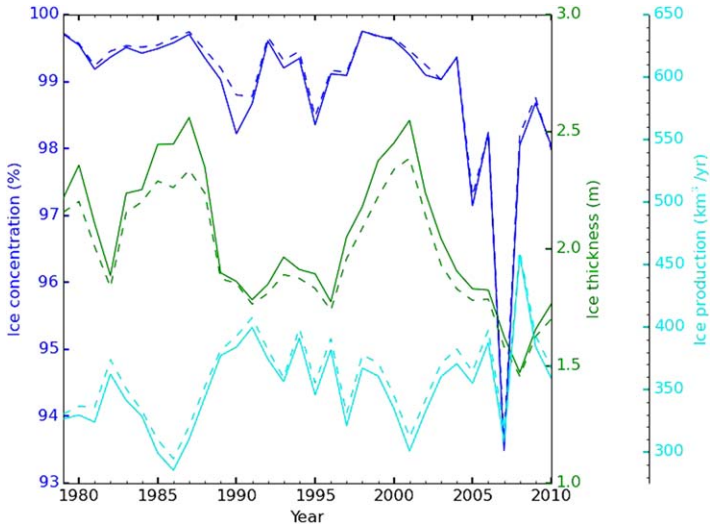

Figure 5. Mean winter (December–April) sea ice time series for the Laptev Sea and East Siberian Sea. CTRL and LF are represented by solid and dash lines, respectively: (a) mean sea ice concentration and thickness with total sea ice production; (b) sea surface salinity of the areas with the production higher than 30 cm, salt flux resulting from the production and river water content.

30 m), while the surface layer at most of the latitudes tends to be fresher than in CTRL. Especially in the surface layer, the differences are not homogeneous and their signs vary geographically. Along with the increased temperature at the top halocline depth this suggests this layer is fed by relatively warm shelf bottom water mixed with brine originating from the sea ice production. This influence is supported by the Hovmöller diagrams of differences at the section across the pathway of the water masses from the East Siberian Sea to the Canadian Basin (Figures 8a and 8b). In LF, this shelf water has continuously larger salinity at the halocline depth (Figure 8a), while the surface layer is typically fresher which is generally associated with higher river water content (Figure 8b). The salinity and river water fraction in the surface layer differences are not constant in time. Notably the salinity difference is positive in the early 1990s and in the mid 2000s. Both events are accompanied with a negative difference in river water fraction. The differences in temperature and salinity described above have consequences for the stratification. The buoyancy frequency N^2 is larger in LF than in CTRL pointing to a higher stability at the winter mixed layer depth.

In contrast, the second section which runs from the East Siberian Sea to Northern Greenland (Figure 9) shows a spatially inhomogeneous pattern in salinity differences that reflects a similar pattern in the river water content differences. While over the shelf the river water content is increased, over the basin it is generally decreased in LF compared to CTRL. At the depth of the halocline, the salinity differences show no clear pattern and suggest no additional feeding from the brine formed in the polynyas in LF, such that the overall stability of the upper halocline is reduced. This is again supported by the Hovmöller diagrams of differences at the section across the pathway of shelf water from the Laptev Sea (Figures 8c and 8d). In LF, the surface layer is mostly more saline, while the time series of the river water fraction differences show a strong seasonal cycle with typically less river water in the surface layer in the winter and more in the summer. The third section runs from the Laptev Sea to the Fram Strait (Figure 10). Here the differences between

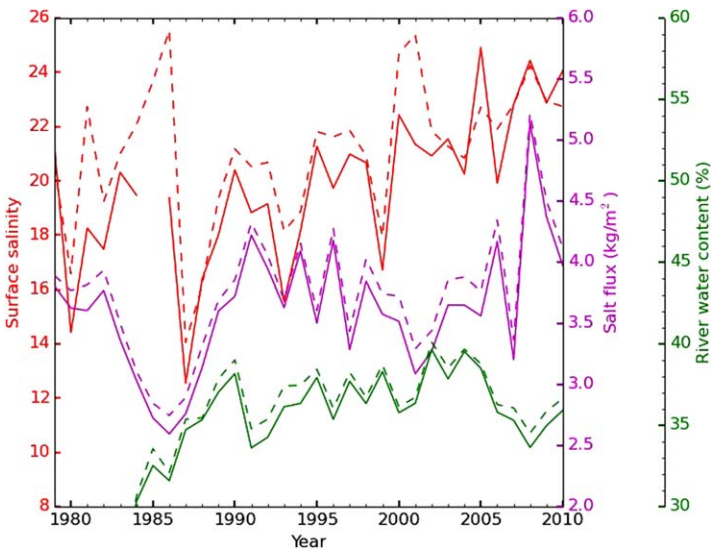

Figure 6. Mean April (2000–2010) buoyancy at the top halocline (25–30 m): (a) CTRL, (b) LF minus CTRL. Black lines mark the profile sections across Canadian, Makarov, and Eurasian Basins (Figures 7, 9, and 10). Gray line marks a section for the time series at the Eurasian shelf break (Figure 8).

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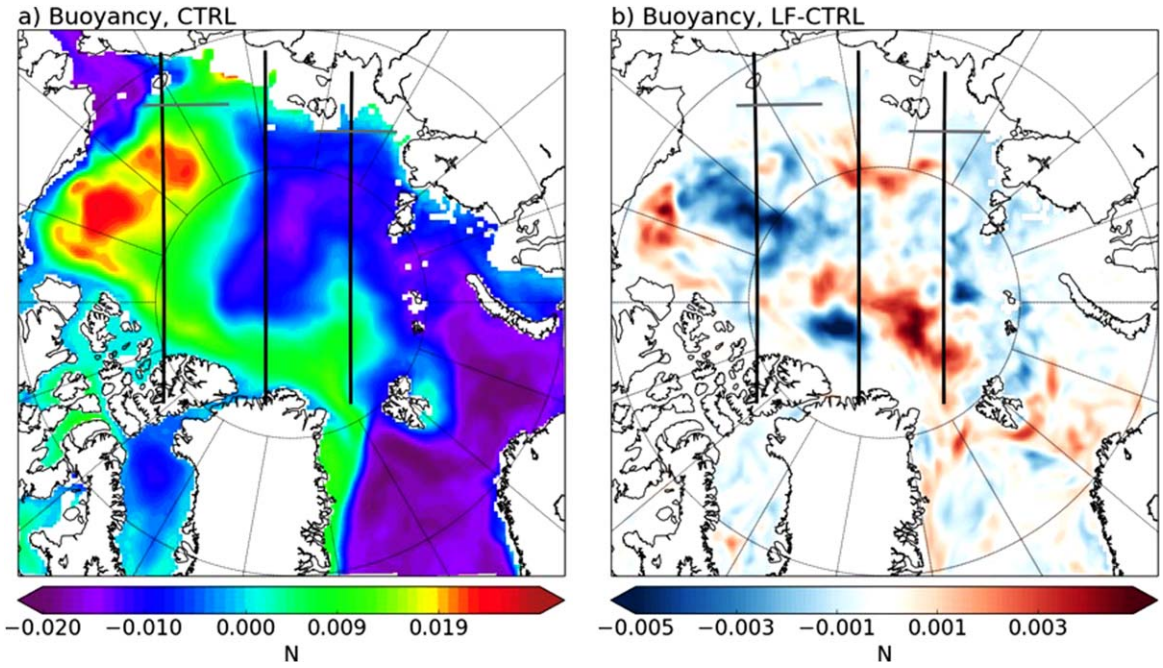

Figure 7. Mean April (2000–2010) salinity, temperature, river water fraction, and buoyancy frequency along the section across the Canadian Basin.

the runs are similar as in the second section, but less pronounced. While there is still less river water in the surface layer in LF than in CTRL, the surface mixed layer is deeper that in the Canadian and Makarov Basins. The halocline is weak and the landfast ice parametrization has no clear effect on it's stability.

5. Discussion

Landfast ice simulated in LF grows almost exclusively thermodynamically and allows very little dynamical sea ice thickening ("growth") due to convergent motion. Therefore, the coastal regions covered by landfast ice in LF are thinner than in CTRL. To some extent, the arbitrary choice of the 25 m isobath for turning on the parameterization places the polynyas in the approximately observed positions. In contrast, in CTRL polynyas are located directly on the coast. The shift of the polynyas from the coast in CTRL toward a realistic landfast ice edge location in LF moves the brine production closer to the shelfbreak and into more saline ocean where the S_{flux} per sea ice volume is increased.

To determine if the amount of brine produced in our simulations is realistic, we compare the simulated sea ice production to the available remote sensing winter sea ice production estimates for the region. There are no such estimates available for the East Siberian Sea, but a number of authors [*Dethleff et al*., 1998; *Willmes et al*., 2011; *Rabenstein et al*., 2013] provide a wide range of estimates for the Laptev Sea. Our simulated sea ice production in the Laptev Sea that covers only about a third for the LESS, amounts to 129 $\pm$ 12km^3 in an average winter in 2000s (the average for the whole LESS is 366 $\pm$ 39km^3). This simulated value falls significantly below the 258 km^3 estimated by *Dethleff et al*. [1998] for winter 1991/1992, but it is larger than *Rabenstein et al*. [2013]'s estimate of 94 $\pm$ 27km^3 for the spatially limited area in the southeastern Laptev Sea for winter 2007/2008. The lowest estimates of the sea ice production in the Laptev Sea are 55 $\pm$ 15 km^3 for the winters 1979–2008 [*Willmes et al*., 2011]; these estimates take into account ice production in areas with sea ice thinner than 20 cm, while in our calculation all the ice production is summed up. Our simulated values are placed in the mid range of the estimates from the remote sensing data and the impact of the brine produced in the simulation is not exaggerated.

Apart from the sea ice production, the river runoff on the shelf seas has an influence on the salinity of the Arctic Ocean surface layer. The pathways of the river water from the Eurasian shelf into the deep basins of the Arctic Ocean are governed by the summer winds [*Guay et al*., 2001; *Dmitrenko et al*., 2005a]. In summers

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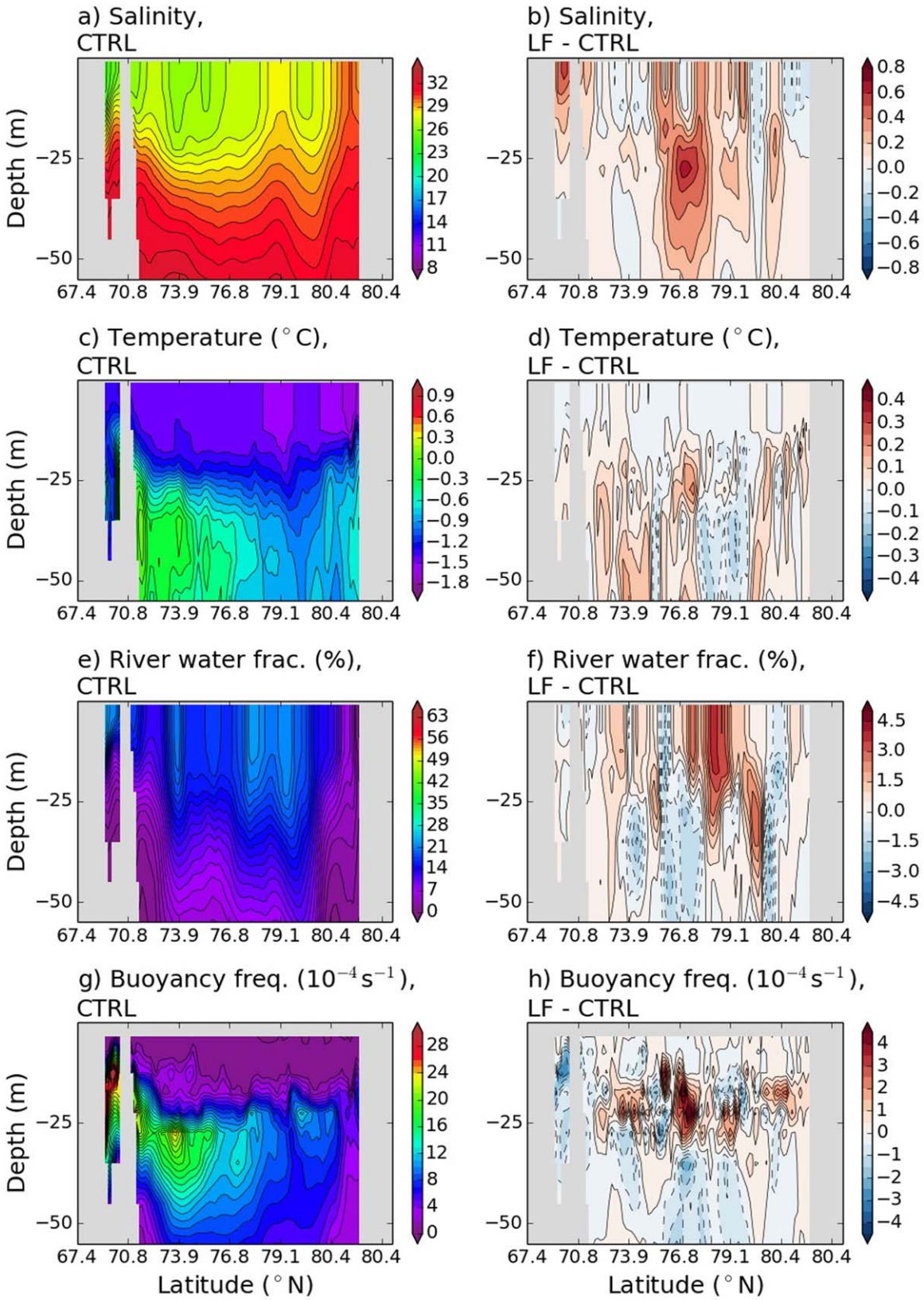

Figure 8. Hovmöller diagrams of monthly mean salinity and river water fraction differences between LF and CTRL: average over an oceanographic section at the Eurasian shelf break north of the East Siberian Sea (a, b) and at the Laptev Sea shelf break (c, d). Sections are marked by gray lines on Figures 1 and 6.

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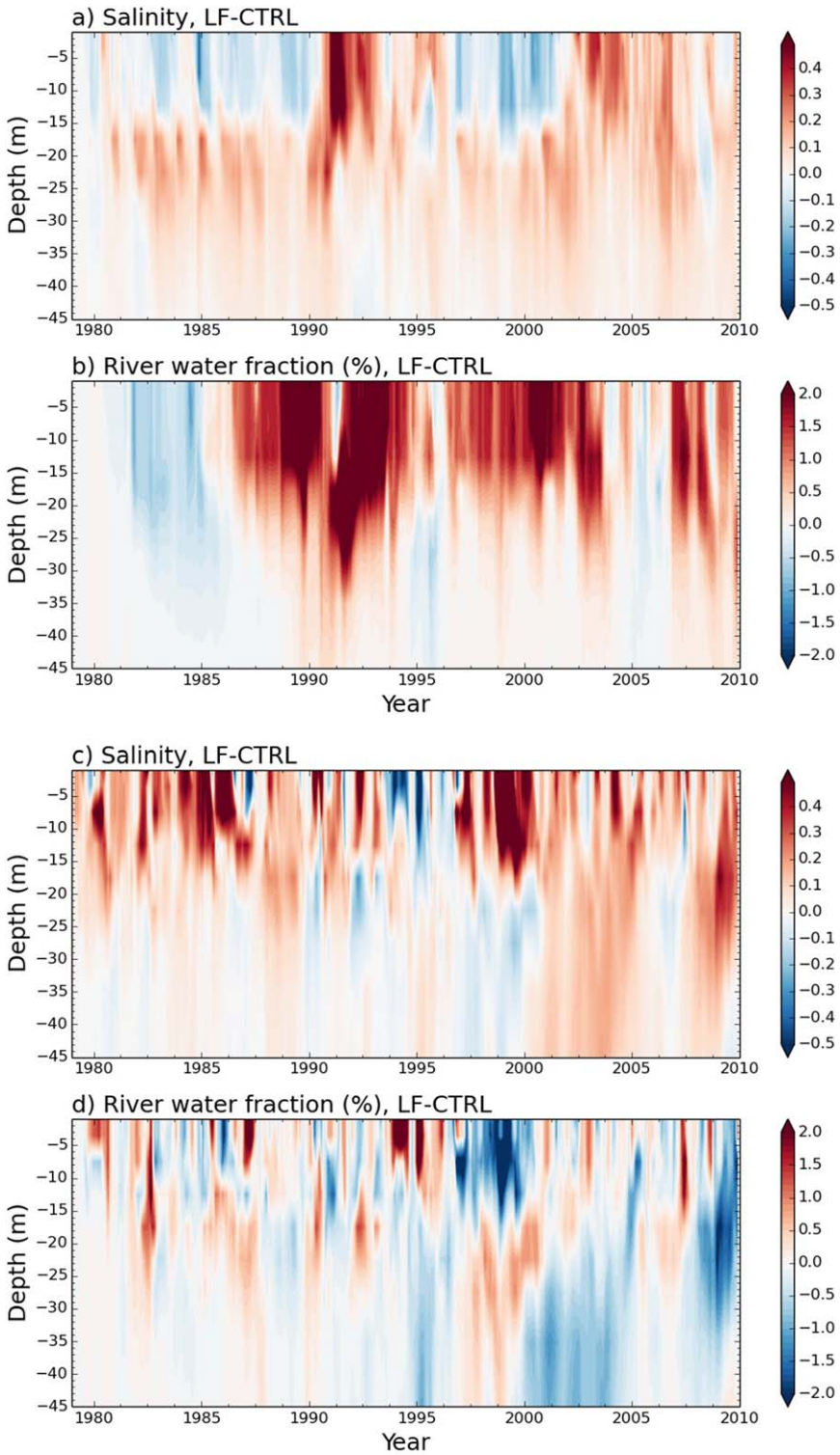

Figure 9. Mean April (2000–2010) salinity, temperature, river water fraction, and buoyancy frequency along the section across the Makarov Basin.

with prevailing southerly winds, the river plume is advected northward into the Eurasian Bains, while in the summers with prevailing northerly winds the plume is stored on the shelf and steered to the East Siberian Sea from where it eventually reaches the Makarov and Canadian Basins. The latter conditions result in high

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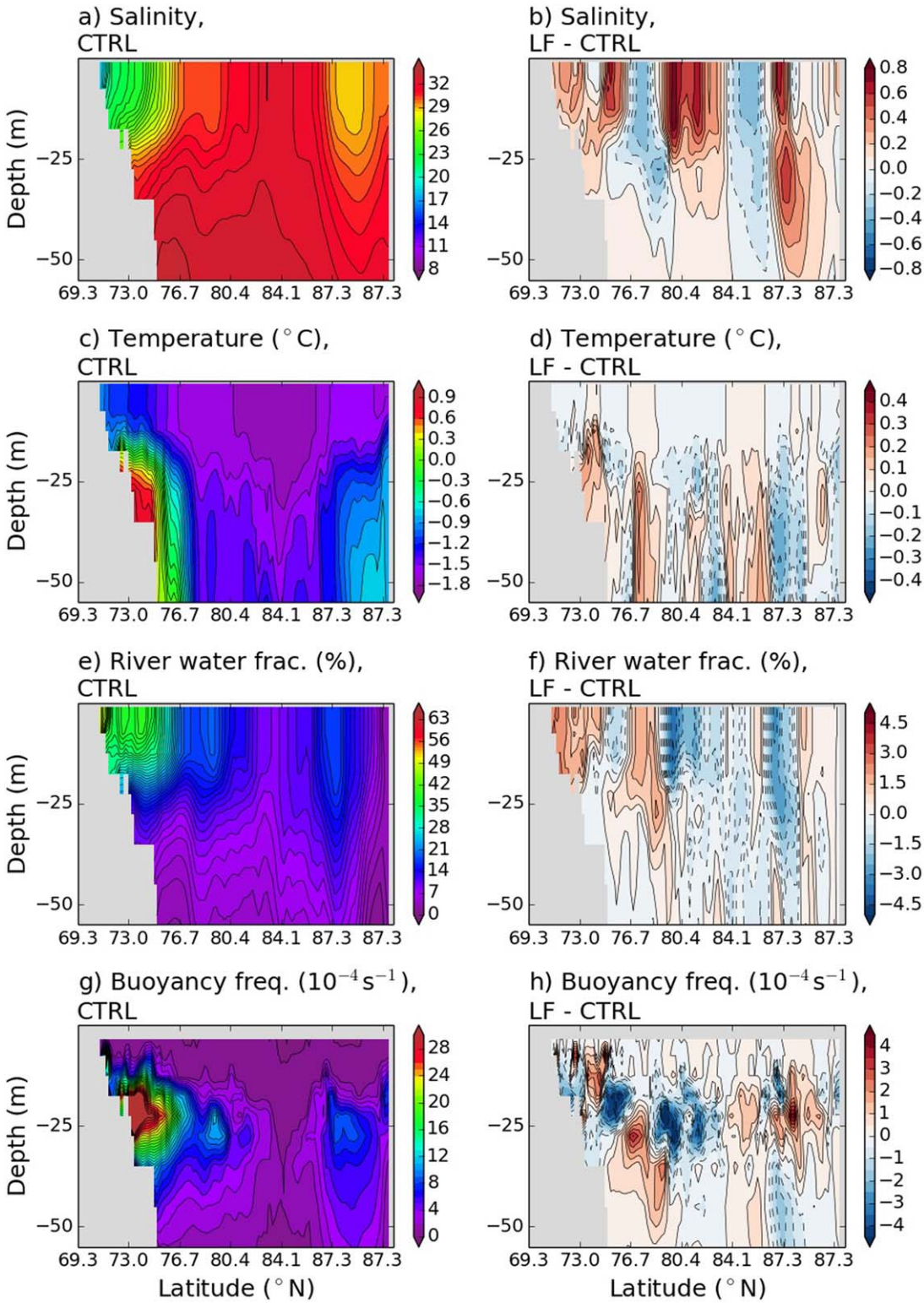

Figure 10. Mean April (2000–2010) salinity, temperature, river water fraction, and buoyancy frequency along the section across the Eurasian Basin.

river water content on the shelf in the winter despite the low amount of the winter runoff of the Siberian rivers. This plume is protected from the winds by an immobile shield of the landfast ice; our LF simulation indicates that with an appropriate representation of landfast ice more river water is stored in the LESS than

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in CTRL where more river water is dispersed by the wind acting through the drifting sea ice. From the LESS, the plume is driven northeastward by geostrophic currents (black arrows on Figure 1) into the Canadian Basin and less river water reaches other parts of the Arctic. The landfast ice breaks up in summer and in LF, in summers with strong southerly winds the increased amounts of the river water stored on the shelf during the winter are driven northwards into the European and Makarov Basins. These occasional summer river water pulses interrupt the generally higher salinity in the surface layer typical for the winter period and contribute to the inhomogeneous salinity and river water fraction differences in the surface layer across the Makarov and Eurasian Basins. Still, the already weak halocline in this sector of the Arctic [*Boyd et al*., 2002; *Rudels et al*., 2004] is further eroded in LF.

Both, the brine formed in the sea ice production and the river water distribution are important for the formation of the Arctic halocline (Figure 2). The Arctic halocline is notoriously difficult to model: in a comparison study, none of the AOMIP models were able to reproduce it [*Holloway et al*. 2007]. The authors suggest that this is a consequence of unresolved physics in the vertical mixing and in the shelf-basin exchange. The deficiency in the vertical mixing scheme physics was addressed by *Nguyen et al*. [2009], who achieved a good representation of the halocline in the Canadian Basin by reducing the vertical background diffusivity and implementing a salt plume parametrization. The latter is efficiently redistributing the brine formed in the sea ice production from the surface layer to their neutral buoyancy depth. Their solution still has a halocline layer in the Makarov and Eurasian Basin that is too warm and too fresh.

In our study, we use the same value of the vertical background diffusivity values as *Nguyen et al*. [2009], but instead of the salt plume parametrization, we apply a landfast ice parametrization thereby addressing some of the unresolved processes in the shelf-basin exchange. Compared to the CTRL, the surface layer in the Canadian Basin in LF freshens due to increased river water content, while the halocline layer becomes more saline due to the difference in the brine production in the LESS polynyas. Although the salinity differences due to the landfast ice parametrization are very minor compared to the effect of the reduced vertical diffusivity (0.1 difference on the basin scale comparing to the differences of 1 in *Nguyen et al*. [2009]), both surface and halocline depth differences make the halocline in the Canadian Basin in LF stronger. Our simulation also supports *Nguyen et al*. [2012]'s conclusions that the Canadian Basin halocline is maintained by the dense water formed on the Eurasian shelf. Additionally, landfast ice parametrization influences the stability of the halocline in the Makarov and Eurasian Basin, where, in contrast to the salt plume parametrization tested by *Nguyen et al*. [2009], the enhanced input of river water in LF freshens and warms the surface layer.

For the broader scientific community, our sensitivity study shows that the landfast ice enhances the stability of the halocline in the Canadian basin and weakens it in the Makarov and Nansen basins. The effect resembles that of the Arctic Ocean salinity anomalies [*Steele and Boyd*, 1998; *Johnson and Polyakov*, 2001; *Morison et al*., 2012] observed in the early 1990s and mid 2002s, where the changes in the wind circulation enhanced the river water export to the Canadian Basin and increased brine production in the LESS polynyas that eroded the Arctic halocline over substantial parts of the Arctic Ocean. Still, during these periods the differences in surface salinity and river water content of the shelf waters originating form the East Siberian Sea change the sign in LF (Figures 8a and 8b). This means that landfast ice is acting as a buffer and has an effect opposing the anomalous offshore winds.

In the future, when less landfast ice cover is expected, this should have implications for the halocline stability. *Yu et al*. [2014] detected a general decrease of the Arctic landfast ice extent of 7 $\pm$ 1.5 % per decade in the period from 1976 till 2007. In the Chukchi, Laptev, and East Siberian Seas also, the landfast ice season is getting shorter, reflecting the general negative trend in the summer sea ice extent in the Arctic shelf seas [*Comiso et al*., 2008; *Stroeve et al*., 2012]. The reduction in the sea ice extent and freezing season duration in the LESS would lead to a weaker halocline in the Canadian Basin, but conversely also to a stronger halocline in the Makarov and Eurasian Basins. Especially for the latter two basins, the Atlantic water layer has been reported to be warming [*Polyakov et al*., 2010]. In such a case, the loss of the landfast ice would inhibit the heat transport from the Atlantic Water layer to the surface and delay the further rapid sea ice loss in the Arctic.

The landfast ice parametrization used in our study strongly depends on the atmospheric temperature through the sea ice concentration (see equation (1)) and can be used in simulations of anticipated climate warming.

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6. Summary and Conclusions

An accurate representation of the landfast ice in a sea ice-ocean coupled model has an impact not only on the winter sea ice and brine production but also on the river water distribution. The landfast ice in the LF shields the river water that remains on the shelf in winter from the winds and promotes northeastward flow of the river water with the coastal currents. We have been able to show this in a sensitivity study with a simple-to-implement landfast ice parametrization that generated landfast ice over extensive areas of the Laptev and East Siberian Seas. In LF, more river water reaches the Canadian Basin and less the Makarov and Eurasian Basin. Also the polynyas are located closer to the shelf break. From there, more brine reaches the halocline depth of the Canadian Basin. Both, the surface salinity decrease due to the river water, and the halocline salinity increase due to the brine rejection both strengthen the halocline in the Canadian Basin, whereas in the Makarov and Eurasian Basins the surface salinity increase erodes the already low halocline stability. Consequently in the latter two basins where the Atlantic water layer is still relatively warm and the halocline is already eroded, extensive landfast ice further weakens the halocline and promotes entraining Atlantic water into the surface mixed layer that might lead to sea ice melt.

Based on our simulations, we recommend to include our landfast ice parametrization to those regional numerical models that address the halocline stability and shelf—deep basin exchanges. The stratification modified by brine produced in the polynyas and the river water that supplies the Arctic Ocean with sediments, nutrients, and pollutants are key factors for biogeochemical processes.

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